

Usage-based Insurance

Using telematics information
for claim prediction

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Contents

- Usage-based insurance
- The distance travelled as the key variable
- The role of driving habits
- Telematics information as complement/substitute of traditional risk factors
- Managing telematics information
- References

Usage based insurance

- UBI (Usage-based insurance):
 - It is a motor insurance where **the premium depends on driving behaviour.**
 - It is a step towards a more **personalized** insurance contract.
 - **Positive effects for the insured:**
 - They will pay a lower premium if they do less kilometers, or drive in a safer way
 - Incentives to drive less and in a safer way
 - It makes insurance more affordable, therefore it seems reasonable to expect that the number of cars without insurance will be reduced.

Usage based insurance

- UBI (Usage-based insurance):
 - **Positive effects for the insurer:**
 - The actuarial accuracy of premiums are improved, the insurer also has a better segmentation of the market.
 - Moreover, companies that offer UBI are viewed as customer-oriented, proactive and environmentally responsible (as PAYD incentives to reduce the use of the car).
 - It can help the company to improve their corporate image and potentially increase its market share.

Usage based insurance

- UBI (Usage-based insurance):
 - **Positive effects for the society:**
 - It contributes to increase traffic safety
 - It contributes to reduce traffic congestion, road maintenance costs, energy consumption and air pollution.
 - There is interest in applying UBI principles in other lines of the insurance business (household, health policies..).

Usage based insurance

- The origin of UBI (Vickrey, 1968):
 - **Pay-At-The-Pump (PATP)**: the driver will pay for his coverage as he buys fuel for the vehicle.
 - **Insured Tires**: an associated insurance company identified in some way with the tire itself, would cover the accident caused by the vehicle using these tires.
- Different UBI pricing options have been proposed:
 - **Per-Mile-Premium (PMP)**: The annual number of kilometers is used as a rating factor into premiums. The insurance company offers a discount for customers driving less than a certain level. This system has been criticized (Litman, 2011) because it is normally based on the driver's self-reported estimate and there is no verification or adjustment at the end of the policy term
 - **Mileage Rate Factor (MRF)**: There is a price per kilometer. Litman (2011) advocates for declining per-kilometer price. The customer will prepay for the number of kilometers he expects to drive during the policy term. There is a verification or adjustment at the end of the policy term based on odometer auditing (there are different proposals to deal with the unused or outstanding kilometers).
 - **GPS-based pricing**: The one implemented nowadays.

Usage based insurance

- **GPS-based pricing:**
 - **Pay-As-You-Drive (PAYD):** Only based on the distance travelled.
 - **Pay-How-You-Drive (PHYD):** It incorporates additional factors related to driving behavior (type of road, part of the day when the vehicle is used, speeding behaviour) into the ratemaking model.
 - **Manage-How-You-Drive (MHYD):** Technology is expected to enable drivers to analyze their competence on the wheel and allow calculating insurance premiums in real time.

The distance travelled as the key variable

Different alternatives on how to introduce the distance travelled in pricing schemes have been discussed, some of them can be found in :

- **Classical vs. generalized offset in GLM vs. GAM:** there it seems to be a non linear relationship between accident rate and distance, although it could be consider a linear relationship between specific ranges of the distance travelled.
- It has also been proposed to use the log of distance in the zero inflated part of a **ZI regression model** in combination with a classical offset, which results in a “modified classical offset” that also has proved non proportionality when applied to real data.

The distance travelled as the key variable

Modification of the classical offset through the introduction of the $\log(\text{distance})$ in the zero inflated part⁽¹⁾:

The Poisson Model with Exposure

When exposure to risk is introduced, then an offset is included in the model. Let us call T_i the exposure factor for policyholder i ($i = 1, \dots, n$), in our case $T_i = \ln(D_i)$, where D_i indicates distance traveled. Then the model can incorporate this factor as follows:

$$E(Y_i | x_i, T_i) = D_i \exp(\beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik})$$

Under this model, the probability of zero using the Poisson distribution is calculated as follows, $P(Y_i = 0) = \exp(-D_i \lambda_i)$, so it depends on the distance and, since λ_i is always positive by definition, then the probability of zero claims declines naturally as distance driven increases.

⁽¹⁾More details in Guillen, M. , Nielsen, J. P., Ayuso, M. and Pérez-Marín, A. M. (2019), “The Use of Telematics Devices to Improve Automobile Insurance Rates”. Risk Analysis 39, 3, 662-672. doi:10.1111/risa.13172.

The distance travelled as the key variable

The ZIP Model with Exposure

Here, we assume that p_i is the probability of an excess of zeros for the i th observation and it is specified as a logistic regression model such that:

$$p_i = \frac{\exp(\alpha_0 + \alpha_1 \ln(D_i))}{1 + \exp(\alpha_0 + \alpha_1 \ln(D_i))}. \quad (1)$$

The Poisson model for Y^* is specified as follows, with an exposure,

$$\begin{aligned} E(Y_i^* | x_i, T_i) &= D_i \exp(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}) \\ &= D_i \lambda_i \end{aligned}$$

$$\begin{aligned} P(Y_i = 0) &= \frac{\exp(\alpha_0 + \alpha_1 \ln(D_i))}{1 + \exp(\alpha_0 + \alpha_1 \ln(D_i))} \\ &\quad + \frac{1}{1 + \exp(\alpha_0 + \alpha_1 \ln(D_i))} \exp(-D_i \lambda_i) \end{aligned}$$

$$\begin{aligned} P(Y_i = k) &= \frac{1}{1 + \exp(\alpha_0 + \alpha_1 \ln(D_i))} \\ &\quad \times (\exp(-D_i \lambda_i)) D_i^k \lambda_i^k / k!. \end{aligned}$$

The distance travelled as the key variable

Using the definition of the expectation of a discrete random variable, the expectation of the Poisson part is:

$$\begin{aligned} E(Y_i|x_i, T_i) &= (1 - p_i) E(Y_i^*|x_i, T_i) \\ &= \frac{1}{1 + \exp(\alpha_0 + \alpha_1 \ln(D_i))} D_i \lambda_i = D_i^* \lambda_i \end{aligned}$$

where

$$D_i^* = \frac{D_i}{1 + \exp(\alpha_0 + \alpha_1 \ln(D_i))}$$

is a transformation of the original exposure D_i .

If $\alpha_1 > 1$, when D_i is large, then D_i^* tends to zero, but when $\alpha_1 < 1$ then D_i^* increases when D_i increases. On the other hand, when D_i tends to zero, D_i^* tends to zero.

If we examine the logistic regression part (Equation (1)), we observe that p_i can be understood again as a transformation of the exposure into the $[0,1]$ interval, which tends to zero when D_i tends to zero if α_1 is positive. Moreover, the derivative of Equation (1) with respect to D_i shows how much the expected claims would change as a function of D_i and indicates that if α_1 is significantly different from zero, then the relationship is not linear.

The distance travelled as the key variable

Zero-Inflated Poisson Model with Offsets (Log of Kilometers per Year in 000s), All Types of Claims

	All Variables		(Only Significant)		Nontelematics		Telematics	
	Coefficient	(p-Value)	Coefficient	(p-Value)	Coefficient	(p-Value)	Coefficient	(p-Value)
Poisson part								
Intercept	-2.148	0.045	-3.396	<0.001	-0.829	0.440	-3.461	<0.001
Age	-0.094	0.232			-0.123	0.121		
Age ²	0.002	0.221			0.002	0.131		
Male	-0.068	0.029	-0.074	0.017	-0.011	0.719		
Age driving license	-0.059	<0.001	-0.056	<0.001	-0.067	<0.001		
Vehicle age	0.014	<0.001	0.014	<0.001	0.017	<0.001		
Power	0.003	<0.001	0.003	<0.001	0.001	0.017		
Parking	0.029	0.420			0.032	0.381		
Log of km per year (thousands) – offset	1.000	–	1.000	–	1.000	–	1.000	–
Km per year at night (%)	-0.004	0.312					-0.001	0.771
Km per year at night (%) ²	0.0001	0.467					0.000	0.931
Km per year over speed limit (%)	0.019	0.001	0.019	0.001			0.018	0.001
Km per year over speed limit (%) ²	-0.001	0.001	-0.001	0.001			-0.001	0.003
Urban km per year (%)	0.026	<0.001	0.026	<0.001			0.027	<0.001
Zero-inflation part								
Intercept (logit)	-0.847	<0.001	-0.857	<0.001	-1.639	<0.001	-0.795	<0.001
Log of km per year (thousands) (logit)	0.404	<0.001	0.410	<0.001	0.824	<0.001	0.406	<0.001
AIC	28,877.112		28,870.556		29,427.423		29,005.172	
BIC	28,999.019		28,951.828		29,508.694		29,070.189	

The distance travelled as the key variable

- The best model is obtained with **classical risk factors** and **telematics information** are considered.
- The **positive sign** of the parameter associated with the log-distance in the logistics part might seem surprising and it could be interpreted erroneously:
 - First of all, it is significant, therefore, it is an evidence of **non proportionality**.
 - This value (0.404) in the first column **does not mean that the greater the distance driven, the greater the probability of the insured having zero claims**.
 - Rather, it means that **the greater the distance driven, the greater the proportion of excess zero claims**, indicating a deviation from the Poisson distribution that can be captured by the ZIP model.

The distance travelled as the key variable

- In the case of the **classical variables**:
 - all the parameters for **gender, driving experience, vehicle age, and the power** of the vehicle are statistically **significant**.
 - Therefore, there is an increasing expectation in the number of claims for women drivers as opposed to men, inexperienced drivers as opposed to experienced, and owners of old and powerful vehicles as opposed to owners of newer and less powerful cars.
- For the **new telematics regressors**:
 - **Two are significant** in explaining the expected number of claims (the percentage of kilometers per year driven over the **speed** limit and the percentage of **urban** kilometers driven per year). Thus, the number of claims increases as these two regressors increase.
 - **No significance** is observed in the case of **night** driving.

The role of driving habits

- The list of variables used (apart from the distance traveled) is very long...
 - Speed
 - Time during the day (night/day)
 - Day of the week (working days/weekend)
 - Location (region, road type)
 - Acceleration/deceleration
 - Turning
 - Contextual information: speed wrt. the rest of drivers, climatological conditions (which is connected to the region where vehicle is used)...

There are many evidences that **these variables are associated to the accident risk.**

The role of driving habits

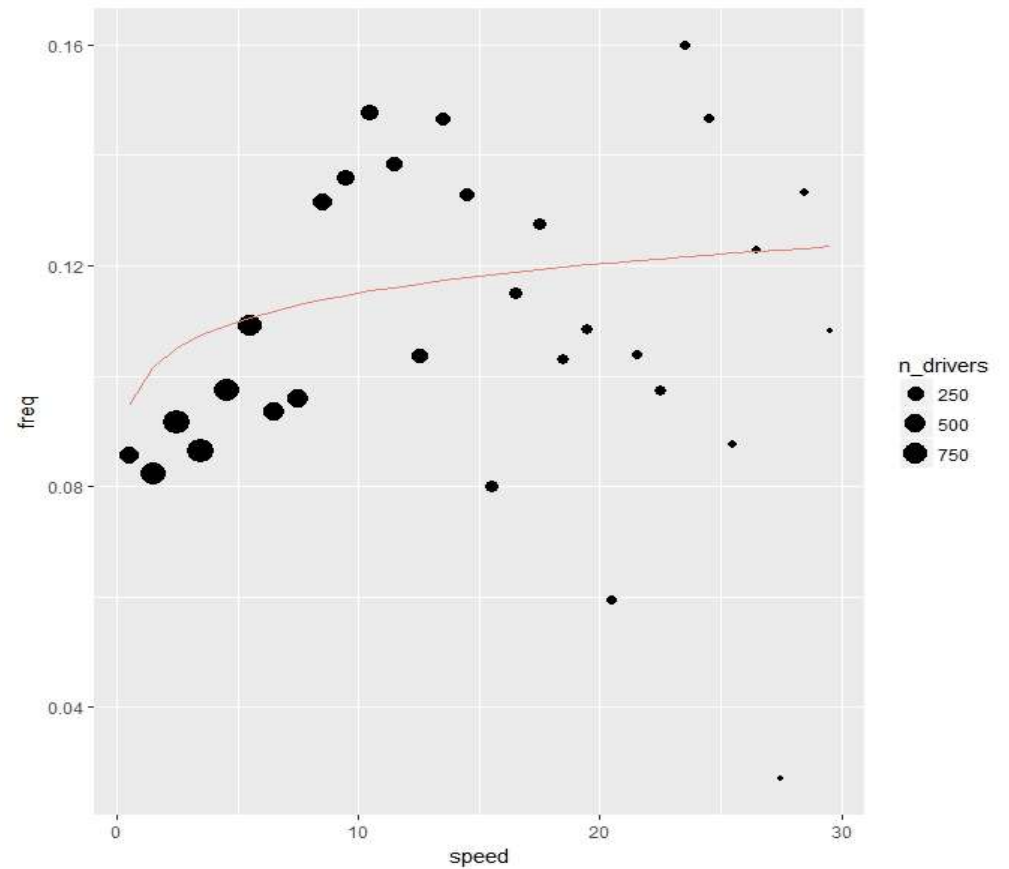
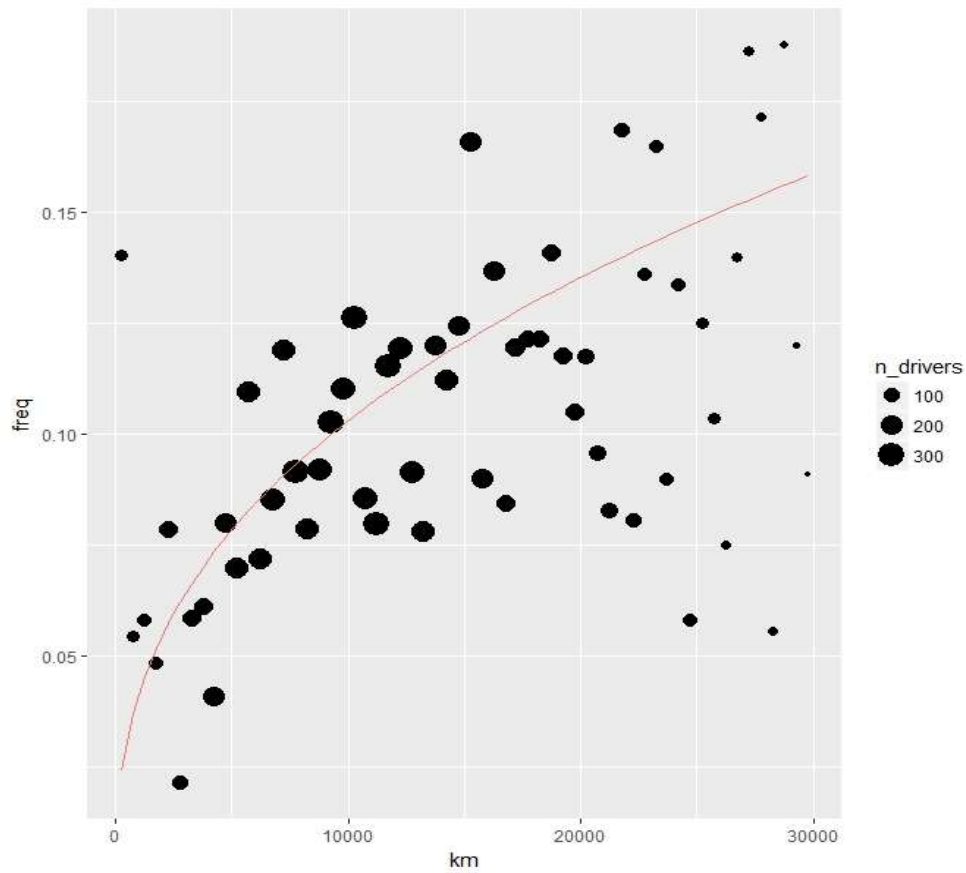
- 9,614 young drivers with a *pay-how-you-drive* (**PHYD**) **policy** in force during the entire year 2011 in a Spanish insurer.

Variable	Definition
km	Distance traveled during the year measured in kilometers
speed	% of kilometers traveled at speeds above the limit
urban	% of kilometers traveled on urban roads
age	Age of the driver
claims	Number of “at fault” accident claims during the year

in Pérez-Marín, A. M. & Guillén, M. (2019) “Semi-autonomous vehicles: usage-based data evidences of what could be expected from eliminating speed limit violations” *Accident Analysis and Prevention* 123, 99-106.

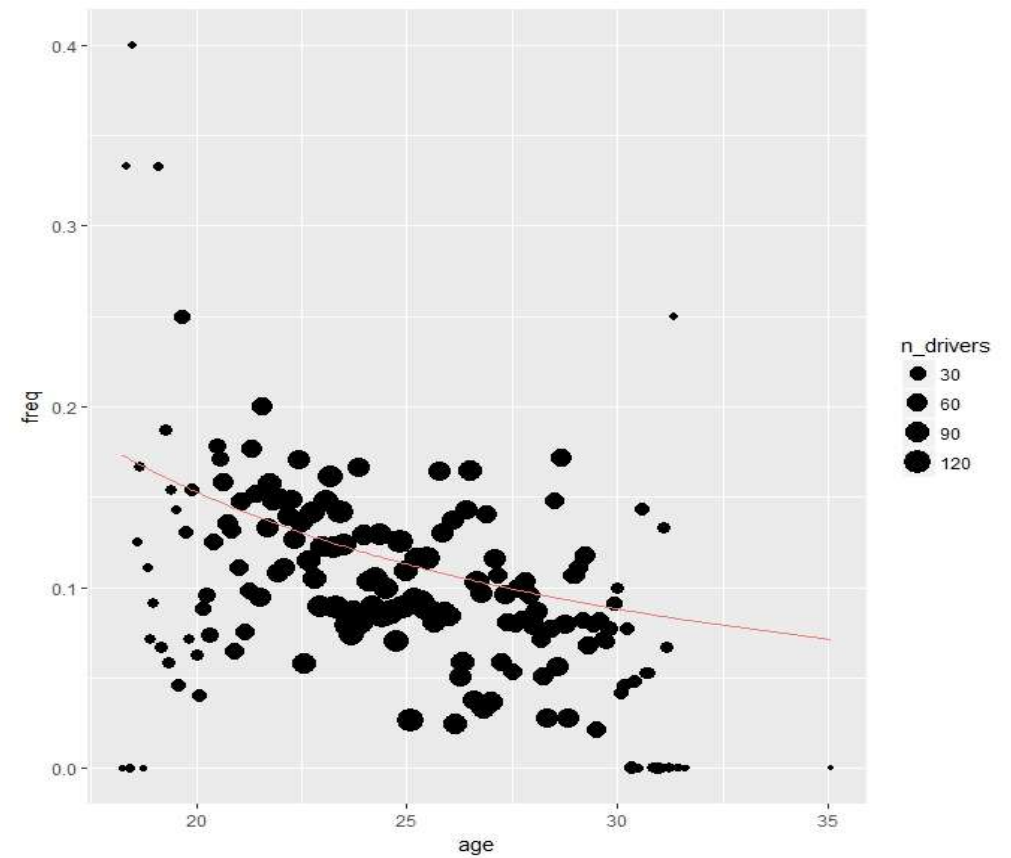
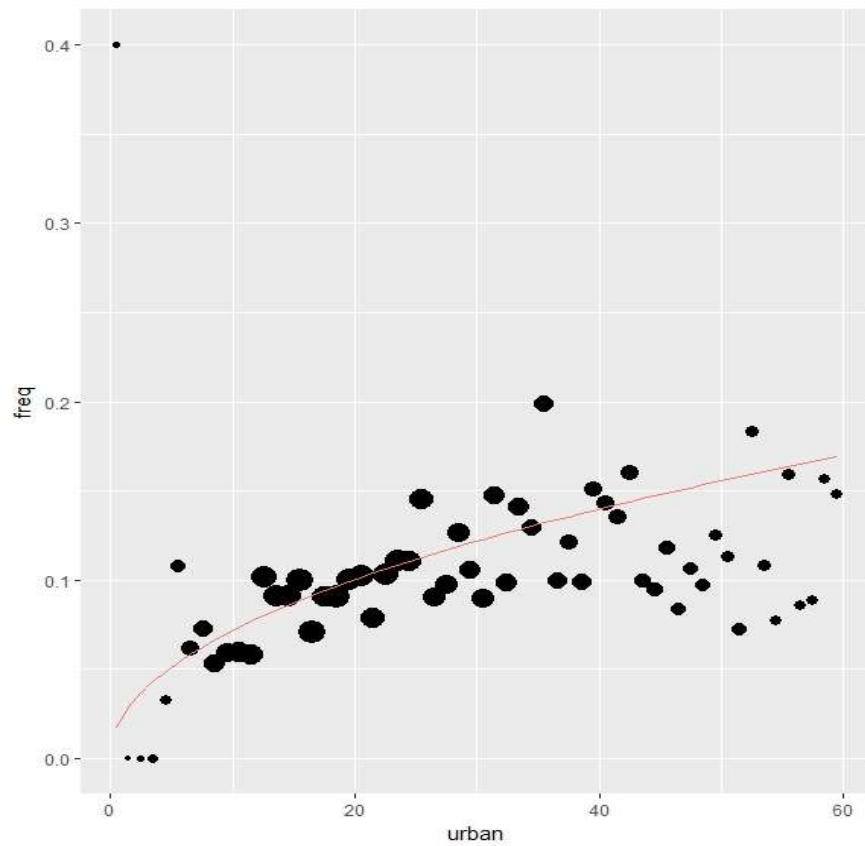
The role of driving habits

Expected claims vs. Km and vs. Speed (%)



The role of driving habits

Expected claims vs. Urban (%) and vs. Age



Telematics information as complement/substitute of traditional risk factors

Telematics as a correction. A two-step procedure:

- Step 1: Let $\hat{Y}_i$ be the frequency estimate obtained as a function of the classical explanatory covariates $x_i = (x_{i1}, \dots, x_{ik})$
- Step 2: Let $\hat{Y}_i^{UBI}$ be the prediction from usage base insurance information $z_i = (z_{i1}, \dots, z_{il})$. Then, let us specify

$$E[Y_i^{UBI} | z_i, \hat{Y}_i] = \hat{Y}_i \exp(\eta_0 + \eta_1 z_{i1} + \dots + \eta_k z_{ik})$$

- The parameter estimates can now be obtained using $\hat{Y}_i$ as an offset.
- In general, telematics information **is a complement**, although when it is used some traditional risk factors in many cases are **redundant**, such as gender.

Managing telematics information

- Managing telematics information for motor insurance is difficult due to its volume and complexity. Information has to be summarized and **key risk factors** identified for pricing purposes.
- **Different approaches** have been proposed to deal with the huge amount of telematics data:
 - It has investigated exactly **how much information should be collected**. Duval et al. (2022) found that telematics data become redundant after about 3 months or 4000 km of records, at least for risk classification.
 - New **variable selection algorithms** have been developed and applied in the context of motor insurance (see Chan et al. 2022).
 - Considering of **near-miss events** in the driving context (an incident that might have resulted in an accident) can help to reduce the amount of information needed for risk classification. Guillen et al. (2020) identified the relationship between telematics information (speed, road type, and time of the day) and three types of near-miss events (harsh acceleration, braking, and cornering).
 - Creating a **driving behaviour score** that will be calculated for each finished trip and help to summarize telematics information.

Managing telematics information

Parameter estimates of the Poisson model of the weekly rate of at-fault claims for the telematics and claims data set

Parameter	Estimate	Standard error	p-Value
Intercept	−8.0637	0.0673	<.0001
EAc1r1	−0.0825	0.0265	.0019
EAc1r2	0.3069	0.1277	.0162
EAc1r3	0.0095	0.0390	.8072
EBrak1	0.0268	0.0086	.0018
EBrak2	−0.4966	0.0770	<.0001
EBrak3	0.0984	0.0336	.0034
EPhon	0.0004	0.0002	.0776
EngineCapacity	0.3644	0.0287	<.0001

More details in Guillén, M.; Nielsen, J. P. y Pérez-Marín, A. M. (2021) Near-miss telematics in motor insurance, *Journal of Risk and Insurance*, 1-21.

Managing telematics information

- **Acceleration events:**

Events of **Intensity 1** have a significant effect and are associated with a lower claim frequency. Events of **Intensity 2** are significantly associated with a higher claim frequency with a positive coefficient. Acceleration events of Intensity 3 do not have a significant effect in the model. We interpret that a **moderate level of acceleration avoids accidents**, while **Intensity 2 is associated with aggressive driving** and thus, with more claims.

- **Braking events:**

Similar conclusion. **Braking events of Intensity 1 and Intensity 3 are associated with a higher claim frequency**, while braking events of **Intensity 2 are associated with a lower claim frequency**. This means that **braking events of Intensity 2 seem to be associated with avoiding at-fault accidents**.

- **Smartphone usage:** It has a significant effect at the 10% significance level ($p = .0776$).

- **EngineCapacity:** It has a positive and significant effect on the claim frequency.

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